IGCSE 0607 Practice Questions — E4.1 — Combined Questions & Markscheme

Coordinates

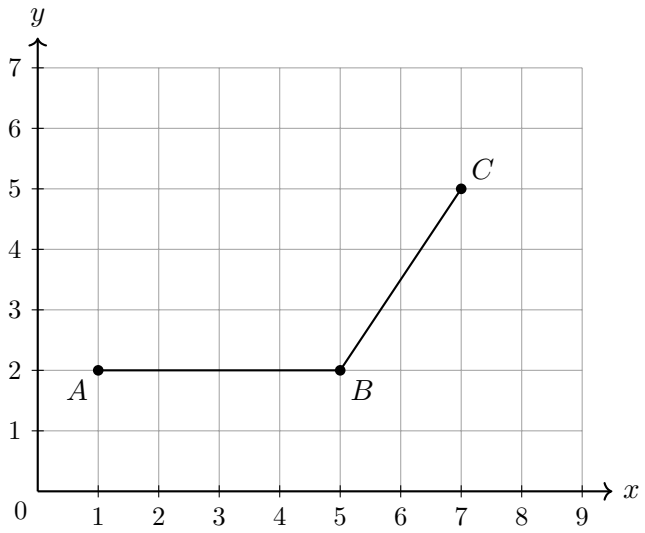
Total Questions: 6 — **Total Marks:** 39 Marks

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1 The diagram shows three vertices, $A(1, 2)$, $B(5, 2)$ and $C(7, 5)$, of a parallelogram $ABCD$.



(a) Find the coordinates of vertex D.

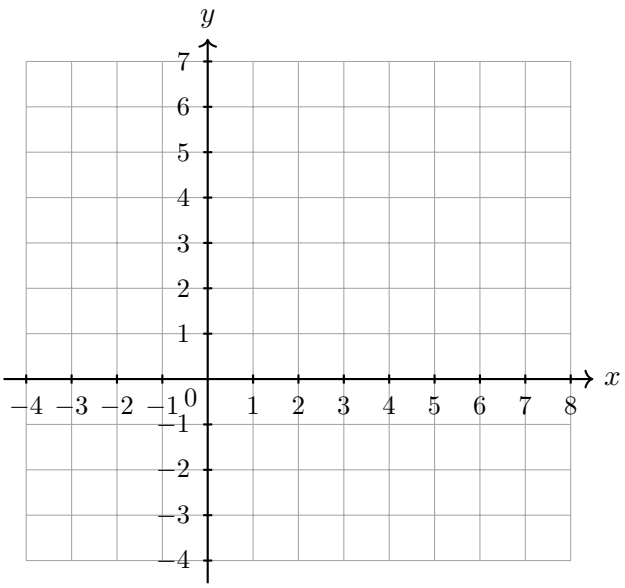
((..... , ...)) **[2]**

(b) Calculate the area of the parallelogram $ABCD$.

..... **[2]**

2 The points $A(-1,3)$, $B(2,-1)$ and $C(6,1)$ are three vertices of a parallelogram $ABCD$.

(a) On the grid, plot the points A, B and C.



[1]

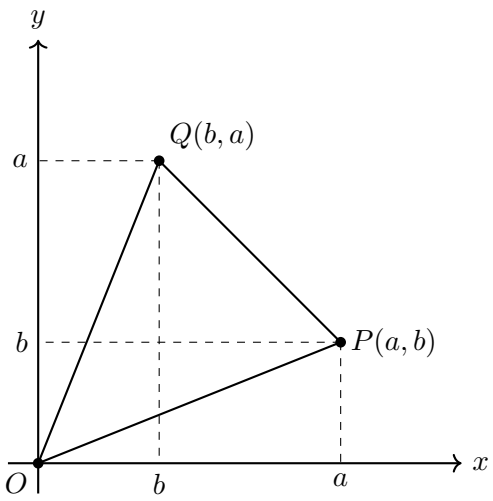
(b) Find the coordinates of the fourth vertex, D.

(..... , ...) **[2]**

(c) Write down the coordinates of the centre of the parallelogram.

(..... , ...) **[2]**

3 The points $P(a, b)$ and $Q(b, a)$ are in the first quadrant, with $a > b$.
The origin is denoted by $O(0, 0)$.



(a) By drawing a suitable rectangle or otherwise, show that the area of triangle OPQ is $\frac{1}{2}(a^2 - b^2)$.

[3]

(b) The area of triangle OPQ is 24 and the length of PQ is $4\sqrt{2}$.
Find the values of a and b.

$a =$ ..... , $b =$ ..... **[5]**

4 A rectangle has its sides parallel to the coordinate axes.
Two opposite vertices of the rectangle are $A(-2, -4)$ and $C(x, y)$, where $x > 0$ and $y > 0$.
The area of the rectangle is 54 square units and its perimeter is 30 units.

(a) Show that $x + y = 9$.

[2]

(b) Form an equation in x and solve it to find the two possible pairs of coordinates for C.

[4]

(c) For the case where $x > y$, write down the coordinates of the other two vertices of the rectangle.

[2]

5 A straight line has the equation $2y + 3x = 12$.
The line intersects the y-axis at A and the x-axis at B.
The point O is the origin $(0, 0)$.

(a) Find the coordinates of A and B.

$A =$ ..... , $B =$ ..... **[2]**

(b) Point C is such that $OACB$ is a rectangle.
Write down the coordinates of C.

((..... , ...)) **[1]**

(c) Point P lies on the line segment AB and has an x-coordinate of 2.
Find the coordinates of P.

((..... , ...)) **[2]**

(d) Find the area of triangle OPB.

..... **[2]**

6 A straight line L has the equation $3x + 2y = 18$.

(a) The line L crosses the x-axis at point A and the y-axis at point B.
Find the coordinates of A and B.

$A =$

$B =$ **[2]**

(b) Point P lies on the line segment AB such that the ratio $AP : PB = 2 : 1$.
Find the coordinates of P.

(..... , ...) **[3]**

(c) Does the point $(4, 3)$ lie on the line L?
Show how you decide.

[2]

1 **(a)** x-coordinate 3 **[B1]**

y-coordinate 5 **[B1]**

(b) Base $AB = 4$ and height $= 3$ **[M1]**

12 **[A1]**

[Back to Question] *[Total Marks: 4]*

2 **(a)** All three points A, B, and C plotted accurately on the grid **[B1]**

(b) Identifies the horizontal and vertical translation, e.g. $\vec{BA} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$ OR correctly plots D on the grid **[M1]**

$(3, 5)$ **[A1]**

(c) Calculates the midpoint of AC or BD: $\left(\frac{-1+6}{2}, \frac{3+1}{2}\right)$ **[M1]**

$(2.5, 2)$ **[A1]**

[Back to Question] *[Total Marks: 5]*

3 **(a)** Identifies the area of the enclosing square as a^2 and the three outer right-angled triangles as $\frac{1}{2}ab$, $\frac{1}{2}ab$, and $\frac{1}{2}(a-b)^2$ **[M1]**

Sets up the subtraction for the area of OPQ: Area $= a^2 - \frac{1}{2}ab - \frac{1}{2}ab - \frac{1}{2}(a-b)^2$ **[M1]**

Correctly expands brackets and simplifies algebraically to $\frac{1}{2}(a^2 - b^2)$ without errors **[A1]**

(b) Sets up the distance squared equation: $(a-b)^2 + (b-a)^2 = (4\sqrt{2})^2$

$2(a-b)^2 = 32$ [M1]

Solves to find $a - b = 4$ (accepting $a > b$) **[A1]**

Uses the area equation:

$$\frac{1}{2}(a^2 - b^2) = 24$$

$a^2 - b^2 = 48$ [B1]

Factorises to $(a-b)(a+b) = 48$, substitutes $a - b = 4$ to obtain $a + b = 12$ **[M1]**

$a = 8$ and $b = 4$ **[A1]**

[Back to Question] *[Total Marks: 8]*

4 **(a)** Width $= x - (-2) = x + 2$ and Height $= y - (-4) = y + 4$ **[M1]**

$2(x+2) + 2(y+4) = 30 \implies 2x + 2y + 12 = 30 \implies x + y = 9$ **[A1]**

(b) Area $= (x+2)(y+4) = 54$ **[M1]**

Substitute $y = 9 - x$ to give $(x+2)(13-x) = 54$ **[M1]**

$x^2 - 11x + 28 = 0$ or equivalent three-term quadratic **[M1]**

$(4, 5)$ and $(7, 2)$ **[A1]**

(c) $(-2, 2)$ **[B1]**

$(7, -4)$ **[B1]**

[Back to Question] *[Total Marks: 8]*

5 **(a)** $A(0, 6)$ **[B1]**

$B(4, 0)$ **[B1]**

(b) $(4, 6)$ **[B1]**

(c) $2y + 3(2) = 12$ **[M1]**

$(2, 3)$ **[A1]**

(d) Base $OB = 4$ and height of $\triangle OPB = 3$ (the y-coordinate of P) **[M1]**

6 **[A1]**

[Back to Question] *[Total Marks: 7]*

6 **(a)** Substitutes $y = 0$ to find $A(6, 0)$ **[B1]**

Substitutes $x = 0$ to find $B(0, 9)$ **[B1]**

(b) Recognises the change in x is -6 and change in y is 9 from A to B **[M1]**

Forms a valid ratio equation for either coordinate, e.g. $x\text{-coordinate} = 6 + \frac{2}{3}(-6)$ OR $y\text{-coordinate} = 0 + \frac{2}{3}(9)$ **[M1]**

$(2, 6)$ **[A1]**

(c) Substitutes $x = 4$ and $y = 3$ into the equation: $3(4) + 2(3)$ **[M1]**

Shows $12 + 6 = 18$ and clearly concludes 'Yes' **[A1]**

[Back to Question] *[Total Marks: 7]*